

Predicting the Effects of a Large-Scale Cash Transfer: Financial Frictions, Occupational Choice, and Local Labor Markets¹

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Abstract

What are the macroeconomic and distributional effects of large cash transfers when implemented at scale? We develop a two-region, small open economy heterogeneous-agent model with occupational choice, financial and real frictions, and labor mobility across regions. We discipline the model with baseline data from a new large-scale experiment in rural Malawi. In the model, transfers generate a sharp increase in consumption and a more persistent expansion in productive expenditures, especially intermediate inputs. There is substantial heterogeneity within and across regions: lower-wealth households exhibit the largest consumption response, while wealthier entrepreneurs account for much of the expansion in capital and intermediate input use. Through general-equilibrium effects, workers in non-treated regions benefit, whereas entrepreneurs there are adversely affected. These effects gradually dissipate as the economy converges back to its pre-transfer equilibrium.

Keywords: Large-scale cash transfers, General equilibrium, Heterogeneous agents, Financial frictions, Occupational choice

1. Introduction

Large transfers to poor households, delivered either as cash or an asset-transfer component of Graduation programs, have emerged as a promising anti-poverty policy [Banerjee et al. \(2015\)](#). Experimental evidence, including long-run follow-ups, shows that such interventions can generate sustained gains in consumption, assets, earnings, and reallocation into higher-return activities, though the magnitude and persistence of these effects vary across settings and program designs². Yet it remains unclear how these gains scale and through which mechanisms they materialize. When transfers are large enough to affect local labor supply and demand, equilibrium wage responses and spillovers to non-recipients may alter their incidence and welfare consequences. As a result, aggregate and distributional effects need not coincide with treatment effects estimated in partial-equilibrium experimental settings. What are the macroeconomic and distributional effects of large-scale cash transfers?

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²See [Bandiera et al. \(2017\)](#), [Banerjee et al. \(2015\)](#) and [Banerjee et al. \(2021\)](#).

How can experimental evidence be used to identify the mechanisms behind persistent poverty and underdevelopment? How does the source of financing shape these effects?

A new large-scale experiment in rural Malawi makes these questions especially timely. GiveDirectly is delivering lump-sum transfers of about \$550 per adult under a design intended to identify spillovers and broader equilibrium effects. We develop a two-region heterogeneous-agent model with occupational choice and labor markets that differ in their degree of integration, then discipline it using baseline data, and finally generate predictions for consumption, wealth, and other variables, with particular attention to how these effects vary across households. The model provides disciplined guidance before endline outcomes are observed, which helps isolate mechanisms, tracing equilibrium channels, and evaluating alternative program designs that are difficult to study from the experiment alone. As experimental evidence arrives, it will provide a sharp benchmark for assessing which margins the model captures well and where additional refinements are needed.

Our model captures several features of the environment in which anti-poverty policies operate in developing countries. Households choose between wage work, subsistence agriculture, and entrepreneurship while facing uninsured productivity risk, stochastic access to wage employment, and tight financial constraints. They can move resources across time by accumulating liquid assets or productive capital, the latter subject to partial irreversibility. This makes entrepreneurship especially risky: if a business fails, capital cannot be fully recovered. We also model intermediate inputs, used in both farming and enterprise production. Importantly, liquidity constraints can limit intermediate-input spending and, with it, the scale of the enterprise. We then embed these household decisions in a two-region small open economy with imperfect labor-market integration, so that a transfer in one region can move wages, reallocate labor, and generate spillovers in the other. This structure allows the model to speak jointly to intertemporal smoothing, production-side choices including occupational reallocation, local price adjustment, and spatial spillovers.

We discipline the model parameters using existing literature and baseline household and enterprise data from Phase I of the GiveDirectly study in Chiradzulu, covering roughly 400 villages across the four southern Traditional Authorities. The data describe a poor, predominantly agrarian setting in which own-food production accounts for the majority of earnings. Market-facing entrepreneurship exists but is small-scale: about one-fifth of households operate an enterprise, most firms employ no paid workers, own savings are the dominant source of start-up finance, and borrowing remains limited even among active entrepreneurs. A household with two adults receives USD 1,100, roughly 69 percent of average annual household consumption in the data. We therefore implement the corresponding model experiment as a one-time increase in liquid assets equal to that share of average consumption in the stationary equilibrium.

Our benchmark experiment implies a temporary aggregate expansion in the treated region, Region 1. Households use the transfer to raise consumption and to finance productive expenditures, but these margins evolve differently over time. Consumption rises sharply on impact and then gradually declines, while capital builds with a lag and remains elevated for longer because part of the transfer is converted into a durable productive stock. The most immediate production response occurs through enterprise intermediate inputs, which rise sharply when the transfer relaxes working-capital constraints. Entrepreneurship also rises meaningfully, by roughly 8 percent of the initial entrepreneur share at the peak. Thus, entry

is an important part of the response, but a substantial share of the productive adjustment occurs through other margins as well: existing and near-margin entrepreneurs expand input use and accumulate capital. The rise in entrepreneurial activity, capital, and input demand increases local labor demand, while entry into entrepreneurship reduces wage-labor supply. Together, these forces push wages in Region 1 up by roughly 15 percent. Temporary labor inflows from Region 2 dampen, but do not eliminate, this wage pressure. Output rises as well, and the boom gradually fades as the one-time transfer is consumed, the additional capital depreciates, and the economy converges back to its pre-transfer stationary equilibrium.

These aggregate responses mask substantial heterogeneity within and across regions. In Region 1, the transfer affects different parts of the wealth distribution through different margins: lower-wealth households exhibit the largest consumption response, middle-wealth households drive most of the entry response, and upper-quartile baseline entrepreneurs account for much of the expansion in capital and enterprise-input use. In Region 2, the untreated region, the impact is smaller and operates through the labor market. As workers move toward Region 1, wages rise temporarily in Region 2, but higher wages reduce entrepreneurial profitability. Entrepreneurship, investment, capital, enterprise inputs, and output decline. Lower- and middle-wealth households receive a brief consumption gain from higher wages, while upper-quartile households, especially baseline entrepreneurs, bear more persistent losses through reduced business profits and capital accumulation.

We also compare the benchmark economy with a fixed-price counterfactual and with alternative degrees of labor-market integration. These exercises show that equilibrium wage adjustment mainly changes the composition and incidence of the transfer response. Holding wages fixed generates a much larger production boom, with stronger increases in entrepreneurship, capital, enterprise inputs, and output. In the benchmark, rising wages dampen this expansion by raising labor costs and the wage-work outside option, but they also shift gains toward non-entrepreneurs and lower-wealth households. Similarly, weaker regional integration amplifies the local wage response and benefits poorer, more labor-dependent households, while stronger integration limits wage pressure and shifts gains toward entrepreneurs and wealthier households. Overall, aggregate welfare in the treated region is similar across these cases; the main differences are distributional.

Finally, we examine how the results change when the transfer must be financed domestically rather than treated as an external inflow. We study a one-time economy-wide transfer financed by government borrowing, with a permanent uniform lump-sum tax used to service the debt. Domestic financing leaves the production response nearly unchanged: output, capital, and intermediate-input paths are almost identical to the externally financed benchmark. The cost instead appears mainly through consumption, which is lower in the long run because households pay the debt-service tax. Welfare remains positive for all wealth quartiles, but aggregate consumption-equivalent gains fall from 10.54 percent under external financing to 8.89 percent under domestic financing. We construct an additional measure to summarize the policy's net value: the wealth-equivalent multiplier. It is defined as the average liquid-wealth value of the policy divided by the transfer size. A multiplier of one means that each dollar transferred is worth one dollar to households; values above one indicate that equilibrium effects amplify the transfer, while values below one indicate that taxes or other equilibrium adjustments erode part of its value. The multiplier is 1.02 under external financing, reflecting modest general-equilibrium amplification, but falls to 0.83 under

domestic debt-service financing.

1.1. Related Literature

Our model builds on a recent macro-development literature that develops quantitative heterogeneous-agent models with financial frictions to evaluate the role of financial markets in accounting for differences in development between countries and over time (Buera et al., 2011; Buera and Shin, 2013; Midrigan and Xu, 2014). This literature finds that differences in financial development are quantitatively important in explaining per-capita income differences. This motivates us to extend and apply these frameworks to analyze the effects of large-scale cash transfers.

More broadly, this paper contributes to the macro-development literature that uses quantitative general-equilibrium models to evaluate large-scale policies in developing economies. Examples include Giné and Townsend (2004) and Kaboski and Townsend (2011), who study financial liberalization and an exogenous microcredit intervention, respectively; Buera et al. (2021), who examine the aggregate and distributional effects of microfinance; and Fujimoto et al. (2023), who analyze the macroeconomic effects of free secondary schooling. Most closely related to our paper is Buera et al. (2014), which studies asset grants targeting the poor in a quantitative occupational-choice model with financial frictions. Our model incorporates capital irreversibility, spatial mobility, intermediate inputs, and agriculture, providing a richer account of how the transfer affects household decisions, local production, and equilibrium prices. More broadly, our paper follows the agenda in Buera et al. (2023) of combining experimental evidence with structural macro frameworks to move from micro treatment effects to economy-wide policy analysis.

This paper also complements the experimental literature on large transfers and Graduation-style anti-poverty programs. Haushofer and Shapiro (2016), Banerjee et al. (2015), Bandiera et al. (2017), and Banerjee et al. (2021) document important direct and medium-run effects of such interventions on treated households, while Banerjee et al. (2022) highlights the difficulty of disentangling mechanisms within bundled programs. At the same time, impacts are not uniform across contexts: Bauchet et al. (2015) finds no lasting average effect in South India, and Barker et al. (2023) documents fading treatment effects in Ethiopia. Most closely related is Egger et al. (2022), which studies the local general-equilibrium effects of large cash transfers experimentally. Our contribution is to bring a quantitative heterogeneous-agent model to these questions, allowing us to trace equilibrium channels, study heterogeneous incidence and welfare, and evaluate counterfactual scale-ups and alternative program designs that are difficult to recover from the experiment alone.

2. Data

The data come from a cash transfer study in Chiradzulu, Malawi (Egger et al., 2024), in which GiveDirectly is implementing a universal, unconditional transfer of US\$550 to each adult. Following Egger et al. (2022), the study design generates treatment variation across geographic units that is intended to identify both direct and spillover effects once the experiment concludes. Our analysis, however, draws only on the baseline data collected before the transfer rollout.

Chiradzulu is a predominantly rural, agrarian district near Blantyre, with approximately 115,000 households and 450,000 residents distributed across 10 Traditional Authorities, roughly 900 villages, and 79 weekly markets. We combine the baseline household and enterprise modules from Phase I of the study, which covers the four southern Traditional Authorities and about 400 villages.

Table 1: Household Summary Statistics

Variable	Mean	Median	SD
<i>Aggregates</i>			
Annual HH earnings (USD)	617.77	416.21	658.76
Income	1.16	0.84	1.12
Earnings per capita (USD)	155.75	—	—
HH members	3.97	4.00	1.71
HH adults (age ≥ 18)	2.18	2.00	1.00
<i>Consumption</i>			
Consumption	2.61	1.91	2.32
Non-durable consumption	2.81	2.14	2.36
<i>Earnings Shares</i>			
Self-employment share	0.14	0.00	0.28
Labor share	0.23	0.00	0.35
Own food production share	0.63	0.78	0.39
<i>Assets</i>			
Productive assets incl. livestock	0.08	0.00	0.22
Gross assets without land	0.32	0.11	0.87
Gross assets incl. land	1.47	0.89	1.93
Net worth	1.45	0.88	1.93

Notes: The sample contains 3,275 households, falling to 3,047 for the earnings-share variables, for which some households are missing the components needed to compute shares. Monetary variables are winsorized at the 2nd and 98th percentiles and normalized by weighted mean household earnings = USD 617.77 unless otherwise stated. Variables denoted in USD are converted from Malawian Kwacha at the market exchange rate prevailing during the survey period (MWK 1,724.13 = USD 1). Non-durable consumption is author-constructed from the official consumption measure by adding imputed rent and subtracting durable-related expenses.

Table 1 reports summary statistics from the household data. Mean annual household earnings are USD 618, or about USD 156 per capita, and households contain roughly four members on average, of whom just over two are adults. Total consumption, which also includes the value of produce retained for self-consumption, equals 2.61 times earnings. Our non-durable measure is somewhat higher, at 2.81 times earnings, because it adds imputed housing rent to the official consumption measure and nets out durable expenditure, a negligible component here. Gaps of this magnitude between measured consumption and measured earnings are not uncommon in rural household surveys, where income from agriculture and

self-employment is difficult to capture accurately. Total income averages 1.16 times earnings and, beyond the earnings components, includes gifts and government transfers. These transfers are especially important for sustaining consumption among households with very low reported earnings.

The composition of earnings underscores the central role of subsistence production in this setting. Earnings comprise own food production, labor income, and other self-employment, which combines non-agricultural business income with crop and livestock profits. Own food production is by far the largest component, accounting for 63% of earnings on average and 78% at the median. Labor income contributes 23%, while self-employment excluding own food production accounts for only 14%. Although own food production is conceptually a form of self-employment, we report it separately because of its quantitative importance and because it differs from more market-oriented enterprise activity. The asset data paint a similar picture of low average wealth, substantial dispersion, and the importance of illiquid assets. Productive assets, including livestock, average only 0.08 times mean household earnings, with a median of zero, indicating that many households hold little directly productive capital. Gross assets excluding land average 0.32 times mean earnings, while gross assets including land rise to 1.47. Net worth is nearly identical at 1.45, implying that land accounts for a large share of household wealth and that debt is modest on average.³

Table 2 partitions households into three groups motivated by the model in the next section. We classify a household as an entrepreneur household if it reports at least one active enterprise, as a worker household if it has no active enterprise but at least one member worked during the past year, and as neither otherwise. These groups account for 21.3, 54.0, and 24.6 percent of the sample, respectively. The ranking in earnings and income is clear: entrepreneur households are the most market-engaged and have the highest resources on average, worker households lie in the middle, and neither households have the lowest earnings and income. Mean household earnings are 1.32 times average sample earnings for entrepreneur households, compared with 1.00 for worker households and 0.72 for neither households; the same ordering appears for total income. Consumption is also highest for entrepreneur households, though the gap between worker and neither households is much smaller than the gap in earnings. The number of adults is very similar across groups, so these differences primarily reflect economic activity rather than household composition.

The composition of earnings also lines up closely with the group definitions. Entrepreneur households derive 43% of earnings from self-employment and 46% from own-food production, while labor income contributes only 11%. Worker households rely much more on labor income, which accounts for 37% of earnings, but own-food production still represents the largest single component at 59%. The neither group is detached from wage work and active enterprise operation by construction. For this group, 90% of earnings comes from own-food production, with a median share of 100%, while the remaining earnings come from non-wage, non-enterprise sources.

Notably, subsistence agriculture is not confined to the neither category but remains important across all three groups. This motivates a model with both market-facing margins,

³Land is valued as the present discounted value of reported rental income using a 6% discount rate, consistent with the interest rate ($r + \delta$) in the model.

Table 2: Group Wise: Summary Statistics

Variable	Mean	Median	SD	N
<i>Panel A: Ent (share = 21.32%)</i>				
<i>Aggregates</i>				
HH earnings	1.32	1.06	1.13	696
Income	1.46	1.19	1.16	696
HH adults (age ≥ 18)	2.28	2.00	1.00	692
<i>Consumption</i>				
Consumption	3.13	2.39	2.50	696
<i>Shares</i>				
Own food production share	0.46	0.47	0.37	689
Self-employment share	0.43	0.34	0.35	689
Labor share	0.11	0.00	0.24	689
<i>Panel B: Worker (share = 54.02%)</i>				
<i>Aggregates</i>				
HH earnings	1.00	0.68	1.05	1,751
Income	1.16	0.85	1.11	1,751
HH adults (age ≥ 18)	2.18	2.00	0.97	1,742
<i>Consumption</i>				
Consumption	2.48	1.84	2.21	1,751
<i>Shares</i>				
Own food production share	0.59	0.69	0.40	1,662
Self-employment share	0.04	0.00	0.16	1,662
Labor share	0.37	0.22	0.39	1,662
<i>Panel C: Neither (share = 24.56%)</i>				
<i>Aggregates</i>				
HH earnings	0.72	0.29	0.96	825
Income	0.88	0.50	1.05	825
HH adults (age ≥ 18)	2.10	2.00	1.05	820
<i>Consumption</i>				
Consumption	2.44	1.68	2.36	825
<i>Shares</i>				
Own food production share	0.90	1.00	0.26	693
Self-employment share	0.10	0.00	0.26	693
Labor share	0.00	0.00	0.00	693

Notes: Monetary variables are winsorized at the 2nd and 98th percentiles and normalized by weighted mean household earnings = USD 617.77 unless otherwise stated.

entrepreneurship and wage work, and a subsistence/own-food margin that remains quantitatively important even for households engaged in market activity. The group-wise patterns therefore discipline both the occupational structure of the model and the calibration targets for market activity and own-food production.

Table 3: Enterprise Statistics

Variable	Mean	Median	SD	N
Annual profit (USD)	868.67	382.80	1371.60	2,309
Value added	-0.29	-0.36	7.02	266
Productive assets excl. land & buildings	0.10	0.00	0.30	2,406
Productive assets incl. land & buildings	1.89	0.02	3.80	2,340
Own savings (dummy)	0.75	1.00	-	2,219
Paid employees (lower bound, capped at 3)	0.23	0.00	0.60	2,406
Business age (years)	8.63	5.31	9.16	2,309

Notes: Annual profit is reported in USD and winsorized at the 2nd and 98th percentiles. Value added and productive assets are normalized by annual profit and winsorized at the 2nd and 98th percentiles. Productive assets including land and buildings value those assets using a 6 percent rental rate. Variables denoted in USD are converted from Malawian Kwacha at the market exchange rate prevailing during the survey period, MWK 1,724.13 per USD. N varies across rows due to missing values.

Table 3 reports summary statistics for active enterprises from the enterprise survey. Enterprise profits are sizable but highly skewed. Mean annual profit is about USD 870, while the median is only USD 383 and the standard deviation is more than USD 1,370, indicating a long right tail. At the same time, these are very small businesses by standard measures. The median enterprise employs no paid workers, and the mean number of paid employees is only 0.23 even when employment is capped at three. Additionally, roughly 85 percent of firms have no paid employees at all, so most enterprises in the sample are effectively self-operated. Sector composition (not shown in table) is also concentrated in low-scale activities: about half of firms are food vendors, underscoring that the relevant enterprise margin in this setting is microenterprise rather than larger-scale formal business activity. Businesses are not especially young, with a median age of just over five years and a mean of 8.6 years.

Asset holdings are highly uneven. Excluding land and buildings, productive assets average only 0.10 times annual profit and the median is zero, indicating that many firms operate with almost no movable productive capital. Including land and buildings raises average productive assets to 1.89 times annual profit, while the median remains only 0.02, revealing a highly skewed distribution driven by a relatively small number of firms with substantial real assets. Financing is also heavily internal: 75 percent of entrepreneurs report using their own savings to start the business, consistent with limited access to external credit. We also construct a measure of value added as revenues minus input costs, but this variable is available for only a small subset of firms and is extremely noisy. As emphasized by [de Mel et al. \(2009\)](#), measuring profits and value added in enterprise surveys is difficult, and that is true in our setting as well. For this reason, reported profits are more informative than value added for describing enterprise heterogeneity and guiding calibration.

3. Model

Building on the literature on occupational choice and financial frictions (see, e.g., Buera et al. (2015)), our model extends the standard framework along several dimensions. First, we add capital explicitly as a state variable, subject to partial irreversibility. This exposes households to investment risk. Second, we introduce subsistence agricultural production as an additional source of income. Third, both agricultural and enterprise production use intermediate inputs, which are subject to a working-capital constraint. Fourth, we incorporate two regions with partial labor mobility across them. We describe the model in detail below.

Time is continuous. An infinitely lived household has access to two assets. The first is a liquid asset b_t , which can be adjusted freely and earns interest rate r . The second is illiquid productive capital k_t , which evolves through investment and depreciation and is subject to partial irreversibility through the adjustment-cost function $\Phi(i_t, k_t)$ defined as follows:

$$\Phi(i_t, k_t) = -(1 - \phi) i_t \mathbf{1}\{i_t < 0\} + \frac{\kappa}{2} \left(\frac{i_t}{k_t + \bar{k}} \right)^2 (k_t + \bar{k}).$$

where $1 - \phi$ gives the fraction of capital lost when it is uninstalled and sold.⁴

The household chooses consumption c_t , investment i_t , and an occupation o_t^s, o_t^w, o_t^e each instant. Here $o_t^s = 1$ denotes full-time subsistence agricultural specialization, $o_t^w = 1$ denotes wage work, and $o_t^e = 1$ denotes entrepreneurship. Occupations are mutually exclusive:

$$o_t^s + o_t^w + o_t^e = 1.$$

Moreover, beyond choosing one active occupation, every household has access to farm production, given by $A_s(m_t^s)^{\gamma_s}(1 + \chi_s o_t^s)^{1-\gamma_s}$. This own-food technology is available to all households regardless of occupation; the specialization premium (χ_s) applies only when the household specializes full-time ($o_t^s = 1$). It captures the subsistence agriculture prevalent in our data. We now describe each of the occupations in detail.

3.1. Occupations

3.1.1. Subsistence agriculture

A household that chooses full-time subsistence agricultural specialization uses the own-food technology with the specialization premium. Its net income, net of current agricultural input purchases, is

$$y^s(b_t) = \max_{m_t^s \geq 0} \{A_s(m_t^s)^{\gamma_s}(1 + \chi_s)^{1-\gamma_s} - p_m m_t^s\} \quad (1)$$

subject to the working-capital constraint

$$p_m m_t^s \leq \lambda_m b_t. \quad (2)$$

where m_t^s denotes subsistence agricultural intermediates such as fertilizers and pesticides, p_m their price, and χ_s captures the premium to full-time agricultural specialization. The

⁴The quadratic term is introduced to aid in the numerical computation of the problem. When calibrating the model, κ is set to a positive but small number.

working-capital constraint captures the requirement that current agricultural input and short-term capital purchases must be paid upfront out of liquid resources—for example, purchases of fertilizer, which are done early in the crop cycle.

3.1.2. Worker

A household that chooses wage work can still operate the non-specialist own-food margin. Its net income is

$$y^w(b_t, x_t; w_t) = x_t w_t + \max_{m_t^s \geq 0} \{A_s(m_t^s)^{\gamma_s} - p_m m_t^s\} \quad (3)$$

subject to the working-capital constraint

$$p_m m_t^s \leq \lambda_m b_t. \quad (4)$$

The wage-access state x_t determines whether the household has access to wage employment. This is meant to capture the fact that wage employment is not always available in our setting.

3.1.3. Entrepreneur

A household that chooses entrepreneurship operates a business, such as cash-crop agriculture or a retail shop, while continuing to have access to the background own-food production margin. Entrepreneurial gross output is given by

$$y_e^g(z_t, k_t, \ell_t, m_t^e) = \bar{c} \left(z_t (k_t^\alpha \ell_t^{1-\alpha})^{1-\nu} \right)^{1-\gamma_e} (m_t^e)^{\gamma_e}, \quad (5)$$

where m_t^e denotes the enterprise intermediate input or short-term capital and γ_e governs its expenditure share in production. The term

$$z_t (k_t^\alpha \ell_t^{1-\alpha})^{1-\nu}$$

corresponds to the standard span-of-control technology widely used in the entrepreneurship literature. We augment this technology with a Cobb–Douglas aggregation of intermediate inputs.

The normalization constant $\bar{c}$ is chosen so that the model nests the standard value-added technology when intermediate inputs are chosen optimally and are not constrained. In particular,

$$z_t (k_t^\alpha \ell_t^{1-\alpha})^{1-\nu} \equiv \max_{m_t^e} \{y_e^g(z_t, k_t, \ell_t, m_t^e) - p_m m_t^e\}.$$

The entrepreneur chooses agricultural input m_t^s , enterprise intermediate input m_t^e , and hired labor ℓ_t . Its net income is

$$y^e(b_t, k_t, z_t; w_t) = \max_{m_t^s \geq 0, m_t^e \geq 0, \ell_t \geq 0} \left\{ y_e^g(z_t, k_t, \ell_t, m_t^e) - w_t \ell_t - p_m m_t^e + A_s(m_t^s)^{\gamma_s} - p_m m_t^s \right\} \quad (6)$$

subject to the joint working-capital constraint

$$p_m (m_t^s + m_t^e) \leq \lambda_m b_t. \quad (7)$$

Entrepreneurs must finance both the current farm input and the current enterprise intermediate input out of the same liquid resources. Therefore, agricultural input use and enterprise input use compete for working capital. A household with little liquid wealth may have a high-return entrepreneurial opportunity but still be unable to purchase the desired amount of enterprise intermediates.

The occupational choice can be written compactly as

$$y(b_t, k_t, z_t, x_t; w_t) = \max \{y^s(b_t), y^w(b_t, x_t; w_t), y^e(b_t, k_t, z_t; w_t)\}.$$

Equivalently, the household chooses occupation indicators (o_t^s, o_t^w, o_t^e) and receives

$$o_t^s y^s(b_t) + o_t^w y^w(b_t, x_t; w_t) + o_t^e y^e(b_t, k_t, z_t; w_t).$$

3.2. Household's Problem

The household solves

$$\max_{\{c_t, i_t, o_t^s, o_t^w, o_t^e\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} \frac{c_t^{1-\sigma}}{1-\sigma} dt \quad (8)$$

subject to

$$db_t = \left[o_t^s y^s(b_t) + o_t^w y^w(b_t, x_t; w_t) + o_t^e y^e(b_t, k_t, z_t; w_t) + r b_t - c_t - i_t - \Phi(i_t, k_t) \right] dt, \quad (9)$$

$$dk_t = (i_t - \delta k_t) dt, \quad (10)$$

$$b_t \geq 0, \quad k_t \geq 0, \quad (11)$$

$$o_t^s, o_t^w, o_t^e \in \{0, 1\}, \quad o_t^s + o_t^w + o_t^e = 1. \quad (12)$$

3.3. Regions and Temporary Migration

We consider an economy with two ex ante identical regions, 1 and 2, each with population share 1/2. A one-time cash transfer is implemented in region 1. Labor markets are only partially integrated: residents may temporarily supply labor in the other region when wage differences make this profitable.

In the initial stationary equilibrium, the two regions are symmetric. Wages, entrepreneurial returns, and occupational composition are identical across regions, so no individual has an incentive to move.

The transfer to region 1 temporarily breaks this symmetry. By raising local labor demand, it may increase wages in region 1 relative to region 2. This creates an incentive for some workers from region 2 to temporarily work in region 1. We assume that temporary movers can work only as wage workers at the destination: they earn the destination wage $w_{1,t}$, but cannot operate an entrepreneurial activity there. This captures the idea that entrepreneurship requires location-specific knowledge, networks, or capital that do not easily move across regions, at least in response to temporary shocks.

Migration is modeled as a reduced-form earnings wedge. Each worker in region 2 is characterized by an idiosyncratic migration cost factor $c \geq 1$, drawn independently of all other characteristics from a Pareto distribution with shape parameter $\theta > 0$:

$$F(c) = 1 - c^{-\theta}, \quad c \geq 1. \quad (13)$$

The factor c summarizes the frictions associated with temporary movement, including travel costs, the disutility of being away from home, and temporary accommodation costs. A worker with cost factor c who works in region 1 receives effective labor income $w_{1,t}/c$. When $c = 1$, the worker faces no mobility friction; when $c > 1$, migration reduces the effective return from destination work.

Residents first choose occupation in their home region. Conditional on being wage workers, region-2 workers then decide whether to supply labor locally or temporarily in region 1. A worker in region 2 earning the local wage $w_{2,t}$ chooses temporary work in region 1 whenever

$$\frac{w_{1,t}}{c} \geq w_{2,t}, \quad (14)$$

or equivalently,

$$c \leq \frac{w_{1,t}}{w_{2,t}}. \quad (15)$$

Since all workers in region 2 face the same local and destination wages, the cutoff $w_{1,t}/w_{2,t}$ is common across them. The fraction of region-2 workers who temporarily move is therefore given by the Pareto CDF evaluated at this threshold.

Let $N_{2,t}^w$ denote the mass of residents of region 2 who choose wage work at time t . This mass is determined by occupational choice within region 2 and may vary over time as local returns change. Temporary movement does not alter residency: workers who temporarily work in region 1 remain residents of region 2 and are still counted in $N_{2,t}^w$.

Let T_t^{mig} denote the mass of region-2 residents who are temporarily supplying labor in region 1 at time t . Then

$$T_t^{mig} = \left[1 - \left(\frac{w_{1,t}}{w_{2,t}} \right)^{-\theta} \right] N_{2,t}^w \quad \text{if } w_{1,t} > w_{2,t}, \quad (16)$$

and $T_t^{mig} = 0$ otherwise. The effective labor supplied in each region is therefore

$$L_{1,t} = N_{1,t}^w + T_t^{mig}, \quad (17)$$

$$L_{2,t} = N_{2,t}^w - T_t^{mig}. \quad (18)$$

In the initial stationary equilibrium, $w_{1,t} = w_{2,t}$, so the threshold equals one and $T_t^{mig} = 0$. No temporary movement occurs.

The parameter θ governs the degree of labor market integration across regions: it determines how responsive temporary migration is to a given wage gap. Low values of θ imply that most workers face large effective mobility costs, so even sizable wage differences induce little movement; high values imply that few workers face such costs, so even small wage differences generate substantial reallocation. In this sense, θ is a reduced-form measure of how integrated the two regional labor markets are.

3.4. Equilibrium

Let $s = (b, k, z, x)$, where $x \in \{0, 1\}$ denotes wage-work availability. The entrepreneur productivity state z follows a continuous-time Markov chain with generator Λ_z , and x follows a continuous-time Markov chain with generator Λ_x . A symmetric stationary equilibrium of

the pre-transfer economy, for given prices (r, p_m) , consists of a wage w , a value function $V(s)$, policy functions

$$c(s), i(s), m^s(s), m^e(s), o^s(s), o^w(s), o^e(s),$$

entrepreneurial labor demand $\ell^e(s; w, p_m)$, and regional stationary distributions G_1 and G_2 over household states such that:

1. *Household optimality.* Given (r, p_m, w) , households solve the following stationary HJB problem:

$$\begin{aligned} \rho V(b, k, z, x) = \max_{\substack{c, i, \\ o^s, o^w, o^e}} & \left\{ \frac{c^{1-\sigma}}{1-\sigma} + V_b(b, k, z, x) \mu_b(b, k, z, x) + V_k(b, k, z, x) \mu_k(b, k, z, x) \right. \\ & + \sum_{z' \neq z} \lambda_z(z, z') [V(b, k, z', x) - V(b, k, z, x)] \\ & \left. + \sum_{x' \neq x} \lambda_x(x, x') [V(b, k, z, x') - V(b, k, z, x)] \right\}, \end{aligned} \quad (19)$$

where

$$\mu_b(b, k, z, x) = o^s y^s(b; p_m) + o^w y^w(b, x; w) + o^e y^e(b, k, z; w, p_m) + rb - c - i - \Phi(i, k), \quad (20)$$

$$\mu_k(b, k, z, x) = i - \delta k. \quad (21)$$

The occupation choices satisfy

$$o^s, o^w, o^e \in \{0, 1\}, \quad o^s + o^w + o^e = 1.$$

The intermediate-input choices m^s , m^e , and hired labor ℓ^e are chosen optimally by households in the static occupation problems defining y^s , y^w , and y^e above.

2. *Stationarity.* The stationary density satisfies the following Kolmogorov Forward Equation. Let $g_j(b, k, z, x)$ denote the density of households in region $j \in \{1, 2\}$. Then, for each (z, x) ,

$$\begin{aligned} 0 = & -\partial_b [\mu_b^*(b, k, z, x) g_j(b, k, z, x)] - \partial_k [\mu_k^*(b, k, z, x) g_j(b, k, z, x)] \\ & + \sum_{z'} \Lambda_z(z', z) g_j(b, k, z', x) + \sum_{x'} \Lambda_x(x', x) g_j(b, k, z, x'), \end{aligned} \quad (22)$$

where μ_b^* and μ_k^* are the drift functions implied by the optimal policies from the household HJB. Each regional density has mass 1/2:

$$\sum_{z, x} \int_{\mathbb{R}_+^2} g_j(b, k, z, x) db dk = \frac{1}{2}, \quad j = 1, 2.$$

3. *Labor-market clearing.* The wage w clears the local labor market in each region:

$$\int o^e(s) \ell^e(s; w, p_m) dG_j(s) = \int x o^w(s) dG_j(s), \quad j = 1, 2. \quad (23)$$

The left-hand side is entrepreneurial labor demand; the right-hand side is effective labor supplied by households who choose wage work.

4. *Regional symmetry.* In the pre-transfer stationary equilibrium, the two regions are identical:

$$G_1 = G_2, \quad w_1 = w_2 = w.$$

Since the wage gap is zero, the migration cutoff equals one and temporary migration is zero:

$$T^{mig} = 0.$$

Once the one-time unexpected cash transfer arrives, the symmetric stationary equilibrium is disturbed. Wages, value functions, policy functions, and the distribution become functions of calendar time t , so the HJB and KFE are modified accordingly along the transition path. Labor-market clearing also incorporates temporary migration across regions, following the migration rule described in Section 3.3.

4. Calibration

This section describes how we discipline the model using micro- and macro-level data and assess its empirical relevance. We distinguish between parameters assigned externally, based on either standard values from the literature or the baseline data, and those calibrated internally to match moments from the Malawian data.

4.1. Externally assigned parameters.

The coefficient of relative risk aversion is set to $\sigma = 1.5$, and the depreciation rate to $\delta = 0.06$, values that are widely used in related quantitative work. We treat the economy as small and open, so the interest rate is taken as exogenous; in the benchmark specification, we set $r = 0$ to capture the lack of saving opportunities (Dupas and Robinson, 2013). For the entrepreneurial technology, we choose $\alpha = 0.33$ and $\nu = 0.2$, which lie in the range commonly used in the literature. p_m is normalized to 1.

The adjustment cost function Φ has 3 parameters, $\phi, \kappa, \bar{k}$. The term $-(1 - \phi)i$ for $i < 0$ introduces partial irreversibility: when an entrepreneur disinvests, she recovers only $\phi \leq 1$ per unit of capital sold. The quadratic term, governed by κ and the small constant $\bar{k}$ are used to ensure smoothness of the problem. We choose $\phi = 0.35$, a relatively low value in the literature (Kermani and Ma, 2023), but present results for the reversible case (in online appendix), which encompasses higher values (Baley and Blanco, 2026). We set $\kappa = \bar{k} = 0.1$ to keep the problem smooth and numerically stable.

To parameterize the labor-market opportunity process, we note that wage employment in Malawi is closely tied to the agricultural season. In a typical year, the main period of labor demand spans roughly October/November through April/May, reflecting labor needs during land preparation, planting, and weeding, with some additional demand during harvest. This is consistent with our data, in which workers report being employed for an average around 5.5 months per year. We therefore set the transition rates governing labor-market access to

$$(\lambda_x^{0 \rightarrow 1}, \lambda_x^{1 \rightarrow 0}) = \left(\frac{12}{6}, \frac{12}{6}\right).$$

The parameter θ governs the responsiveness of temporary migration to wage differences. We calibrate it from Bryan et al. (2014), who study seasonal migration in rural Bangladesh.

In their control group, 36.0% of eligible households send at least one seasonal migrant, and the paper reports a descriptive earnings comparison of roughly 100 Tk/day for migrants at destination versus 65 Tk/day for a sub-sample of non-migrants in salaried work at origin. Imposing these moments on our migration rule yields $\theta \approx 1.04$, which we take as a reduced-form benchmark for migration responsiveness. We also report how results vary with θ in section 6.2.

The parameter λ_m governs the amount of market input expenditure a household can finance out of liquid wealth. The data suggest that external borrowing is limited and own savings are the dominant source of funding. Only 34% of firms report being confident they could raise one month of revenue by the following month for investment, and just 33% report any positive loan balance. Outstanding debt is 0.46% of total enterprise revenue (annual) in the full sample, rising to 2.5% when restricted to borrowers. These facts motivate a conservative baseline with $\lambda_m = 1$.

Table 4: Externally Set Parameters

Parameter	Description	Value
σ	Risk aversion	1.5
δ	Depreciation rate	0.06
r	Interest rate	0.00
p_m	Normalization	1
ϕ	Partial irreversibility	0.35
$\kappa, \bar{k}$	Quadratic adjustment cost	0.1, 0.1
α	Production function (Capital)	0.33
ν	Span of control	0.2
$\lambda_x^{0 \rightarrow 1}, \lambda_x^{1 \rightarrow 0}$	Wage opportunity process	2, 2
θ	Migration parameter	1.04
λ_m	Working Capital Liquidity	1

4.2. Internally calibrated parameters

The remaining parameters, A_s , γ_s , γ_e , χ_s , ρ , and the z_t process, are calibrated jointly by matching a set of model-generated moments to their empirical counterparts.

The discount rate ρ is chosen to match the aggregate net-worth-to-earnings ratio. The data value is 1.45, and the calibrated model matches this moment with $\rho = 0.15$.

We discipline γ_s , the elasticity of subsistence agricultural output with respect to intermediate inputs, using agricultural input spending relative to own-food production. Agricultural inputs are measured as spending on fertilizer, seeds, and pesticides. The data moment is 0.084. Since liquidity constraints suppress realized input use, matching this low expenditure share requires a slightly higher technological input elasticity, $\gamma_s = 0.09$.

The subsistence agricultural productivity parameter A_s is chosen to match the aggregate own-food share, defined as total own-food production divided by aggregate household earnings. This ratio is 0.65 in the data. Matching this moment gives $A_s = 1.35$.

χ_s governs the payoff from specializing in subsistence agriculture and helps determine the mass of households outside wage work and entrepreneurship. We choose χ_s to match the 0.25 share of households in the neither group, which gives $\chi_s = 0.02$.

Table 5: Internally Calibrated Parameters

Parameter	Description	Value	Targeted moment	Data	Model
ρ	Discount rate	0.15	A/Y	1.45	1.47
σ_ϵ	Productivity dispersion	0.73	Entrepreneur mean/median	2.27	2.25
χ_s	Subsistence premium	0.02	Subsistence agri share	0.25	0.25
γ_e	Ent intermediate share	0.78	Enterprise intermediates / revenue	0.65	0.63
A_s	Agri productivity	1.35	Own-food / agg. earnings	0.65	0.67
γ_s	Agri input elasticity	0.09	Agri inputs / own-food	0.08	0.08

The parameter γ_e governs the importance of intermediate inputs in entrepreneurial gross output. We discipline it using the enterprise intermediate-expenditure share. In the data, input expenditure is monthly spending on stock and inventory for resale, relevant for traders and vendors such as food stalls, groceries, and retail sellers, plus materials and supplies used in operation, relevant for production and service businesses such as tailors, carpenters, restaurants, and salons. Dividing this measure by same-period revenue gives an expenditure share of 0.65. The calibrated value is higher, $\gamma_e = 0.78$, because the model expenditure share need not equal the technological input share. Many entrepreneurs are constrained by the working-capital constraint $p_m m_e \leq \lambda_m b$, which reduces realized input purchases relative to the unconstrained optimum. Matching the observed input-expenditure share therefore requires a higher value of γ_e .

We model entrepreneurial productivity in logs, $\tilde{z}_t \equiv \log z_t$, as a four-state continuous-time Markov chain. The chain is designed to provide a parsimonious approximation to a latent AR(1) process in logs,

$$\tilde{z}_{t+1} = \rho_z \tilde{z}_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2). \quad (24)$$

As a discretization choice, we fix the invariant mass vector ex ante at

$$p = (0.25, 0.25, 0.25, 0.25),$$

so that each state carries one quarter of the stationary mass. Given $(\rho_z, \sigma_\varepsilon)$, we assign one representative grid point to each quartile and scale the resulting support to match the stationary dispersion implied by the AR(1). We then normalize productivity levels so that $\mathbb{E}[z] = 1$. Finally, for this fixed support and invariant mass vector, we recover the generator λ_z so that the chain has stationary distribution p and reproduces annual persistence ρ_z . Thus, in the discretization, σ_ε governs the spread of the productivity grid, while ρ_z governs the persistence of movements across states.

Because our data are cross-sectional, they are uninformative about persistence. We therefore set $\rho_z = 0.85$ following the external evidence in [Asker et al. \(2014\)](#), and choose σ_ε to match the mean-to-median ratio of entrepreneurial profits, which equals 2.27 in the data.

5. Benchmark Results

This section presents the benchmark experiment. We begin by describing how the cash transfer is implemented in the model. We then organize the results by region. We first study the direct effects in Region 1, the treated region, combining the aggregate dynamics with the distributional responses across pre-transfer wealth quartiles and baseline entrepreneur status. We then turn to Region 2, which does not receive the transfer but is partially connected to Region 1 through the labor market, and study both its aggregate spillovers and the household-level heterogeneity behind them.

5.1. Benchmark Experiment

As noted earlier, the cash transfer is \$550 per adult. With 2.18 adults in the average household, this amounts to about \$1,199 per household. In the baseline data, this is about 69% of mean household non-durable consumption. While the transfer is also large relative to reported household earnings ($\times 1.94$), income is measured with considerable noise in this setting, so we view consumption as more reliable benchmarks for scaling the intervention. Our model is formulated at the household level and abstracts from children and intra-household allocation, so we map the adult-level intervention into a single lump-sum household transfer. We set the transfer (Tr) equal to 69.1% of mean household consumption in the stationary equilibrium, matching the empirical ratio. Formally, the transfer is modeled as a one-time, unanticipated increase in the liquid assets. Each agent's state transitions as

$$(b, k, z_i, x_i) \longrightarrow (b + Tr, k, z_i, x_i)$$

where Tr denotes the transfer amount.

5.2. Treated-Region Effects

Figure 1 shows the aggregate response in the treated region. Liquid wealth rises on impact by about 67 percent of steady-state output. Households use the transfer both to raise current consumption and to finance productive expenditures: consumption increases by more than 20 percent on impact, investment jumps by about 6 percent of steady-state output, and intermediate-input spending rises sharply. The input response is almost entirely driven by enterprise intermediate inputs, M_e . Subsistence-agricultural inputs, M_s , remain essentially unchanged, peaking at less than 1 percent. In the calibrated model, most households are not liquidity constrained on the subsistence-agriculture margin, so the transfer does not generate a large increase in agricultural input purchases. Liquid wealth is then drawn down quickly, but total wealth declines more slowly because part of the transfer is converted into productive capital.

The capital stock adjusts gradually, peaking several years after the transfer at just under 5 percent of steady-state output. Entrepreneurship also rises meaningfully: the entrepreneur share increases by almost 3 percentage points at its peak, roughly 8 percent of its baseline level. This entry, or extensive margin matters, but it is not the main source of the aggregate production response. The larger effect comes through the intensive margin: the transfer relaxes liquidity and working-capital constraints for existing and near-margin entrepreneurs, allowing them to scale up through intermediate purchases and capital accumulation. This expansion raises labor demand, while the rise in entrepreneurship reduces local wage-labor

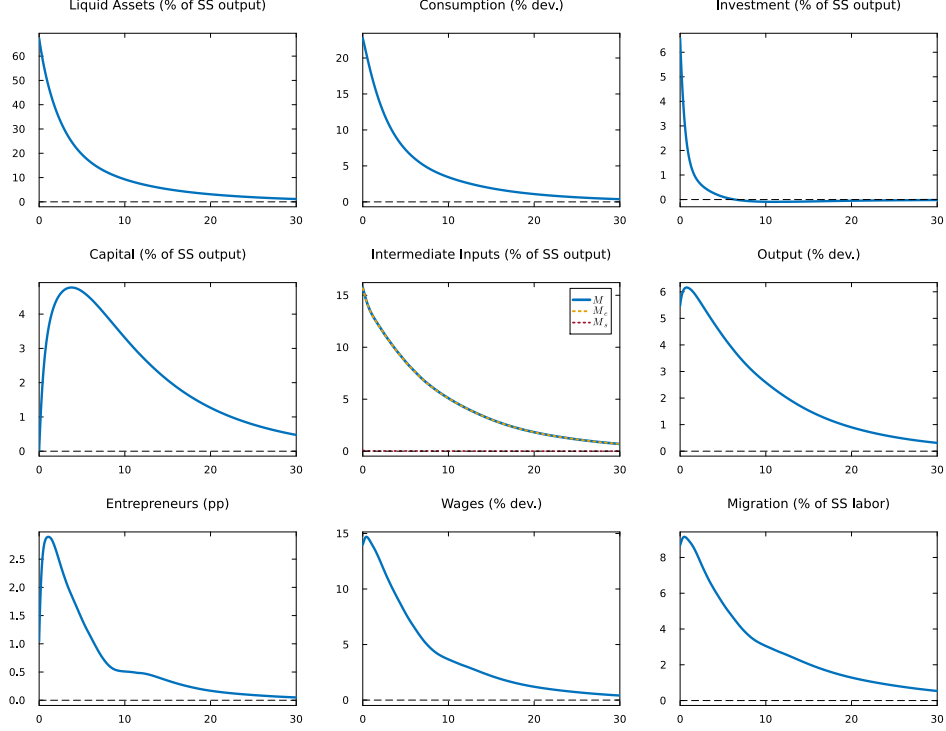


Figure 1: Aggregate Dynamics: Treated region

Notes: The figure shows dynamic responses of aggregate variables in Region 1, the cash-transfer region, under the partially integrated benchmark. Liquid wealth, investment, capital, and intermediate inputs are expressed as deviations from their steady-state values, normalized by steady-state output. Consumption, output, and wages are percent deviations from steady state. Entrepreneurship is reported in percentage points, and migration is measured as a percent of steady-state labor. The intermediate-input panel decomposes total intermediate spending into enterprise inputs, M_e , and subsistence agricultural inputs, M_s .

supply. Together, these forces push wages up by roughly 15 percent. Partial integration with Region 2 induces net in-migration, which dampens but does not fully offset the local labor-market tightening. Output rises by about 6 percent initially and remains above baseline while the added capital and liquidity persist. Because the transfer is one-off, these effects eventually dissipate as liquid assets are consumed, the additional capital depreciates, and households are reshuffled by idiosyncratic shocks.

Figure 2 studies the heterogeneity behind the aggregate responses by decomposing the effects across pre-transfer wealth quartiles and by the contribution of households that were already entrepreneurs at baseline. All responses are measured relative to the counterfactual path of the same group without the transfer. The consumption response is largest in the bottom half of the wealth distribution and declines with wealth: poorer households use a larger share of the transfer to raise current consumption, while wealthier households spread the additional resources more gradually over time. The productive response has the opposite pattern. Capital accumulation and enterprise-input purchases are small in the bottom quartile, larger in the second quartile, and concentrated especially in the third and top quartiles. The yellow dashed lines show that much of this upper-quartile response comes from house-

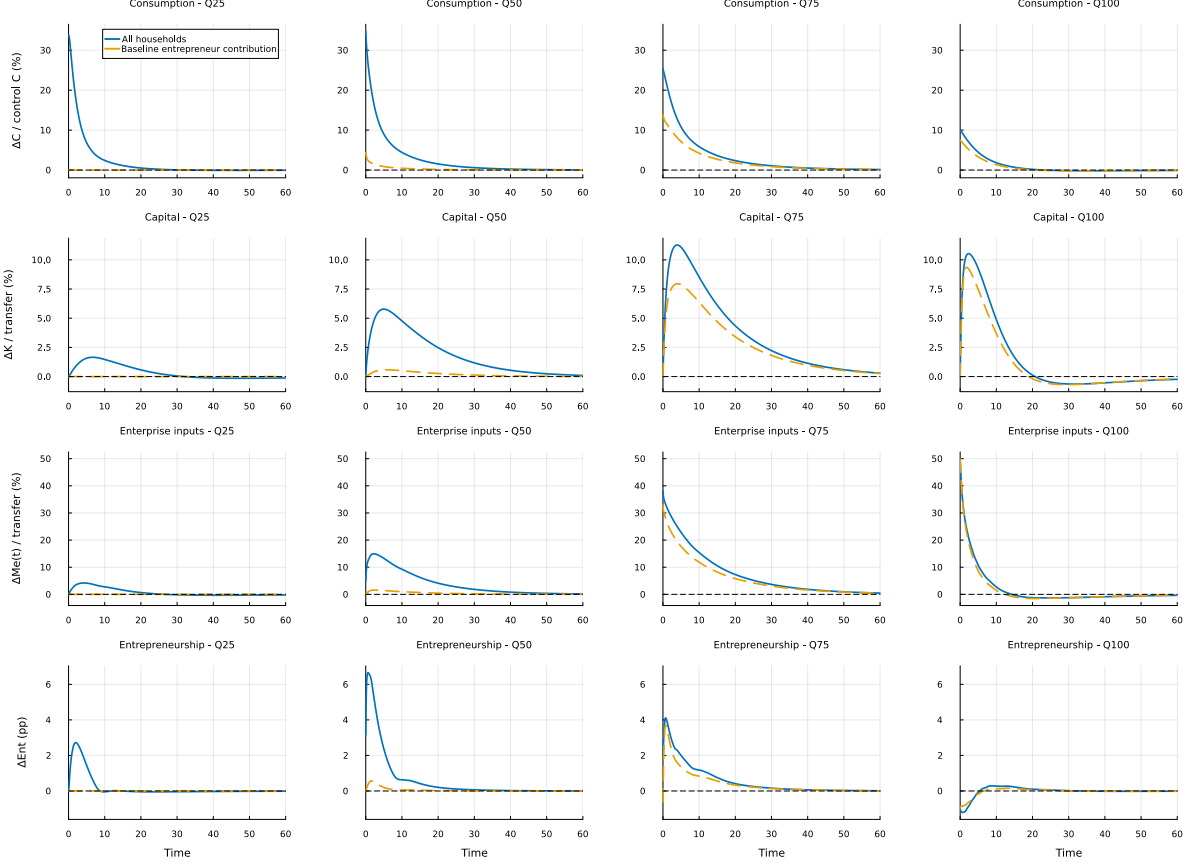


Figure 2: Treated Region Heterogeneous Responses

Notes: The graph shows responses by pre-transfer wealth quartile. Blue lines show total quartile responses; yellow dashed lines show the contribution of baseline entrepreneurs. Each response is relative to the same group without the transfer. Consumption is in percent; capital and enterprise inputs are percent of the transfer; entrepreneurship is in percentage points.

holds that were already entrepreneurs before the transfer. This suggests that the transfer operates partly on the intensive margin: even among upper-quartile households, baseline entrepreneurs appear to face liquidity or working-capital constraints that keep them below their unconstrained scale of operation.

The entrepreneurship response reveals a different margin. The net increase in entrepreneurship is strongest in the middle of the wealth distribution, especially the second quartile. In the third quartile, however, the response is driven largely by baseline entrepreneurs, suggesting more retention or continuation in entrepreneurship rather than new entry. In the top quartile, entrepreneurship initially falls slightly and then rises only modestly. This pattern is consistent with financing frictions being most relevant for high-productivity households that are productive enough to operate a business but do not have enough liquidity to enter or remain active. The transfer relaxes this constraint most strongly for households near the entrepreneurship cutoff, rather than for the poorest households or for the already wealthy.

Figures 1 and 2 therefore tell a consistent story. Poorer households account for much of the immediate consumption response. Middle-wealth households account for much of the increase in entrepreneurship, while in the upper-middle quartile part of the entrepreneurship

response reflects greater continuation among baseline entrepreneurs rather than new entry. Wealthier baseline entrepreneurs account for a large share of the expansion in capital and enterprise inputs. Thus, entry is an important part of the response, but a substantial share of the productive adjustment occurs through other margins as well: existing entrepreneurs scale up input use and accumulate capital.

5.3. Spillovers to the Untreated Region

We now turn to Region 2, the untreated region. Figure 3 reports the aggregate response there. Although households in Region 2 do not receive the cash transfer, they are affected through the partially integrated labor market. The transfer generates a boom in Region 1, raising wages there and inducing temporary worker migration out of Region 2. This outflow reaches almost 9% of Region 2's steady-state labor on impact. The decline in local labor supply tightens Region 2's labor market and raises wages by more than 4% on impact.

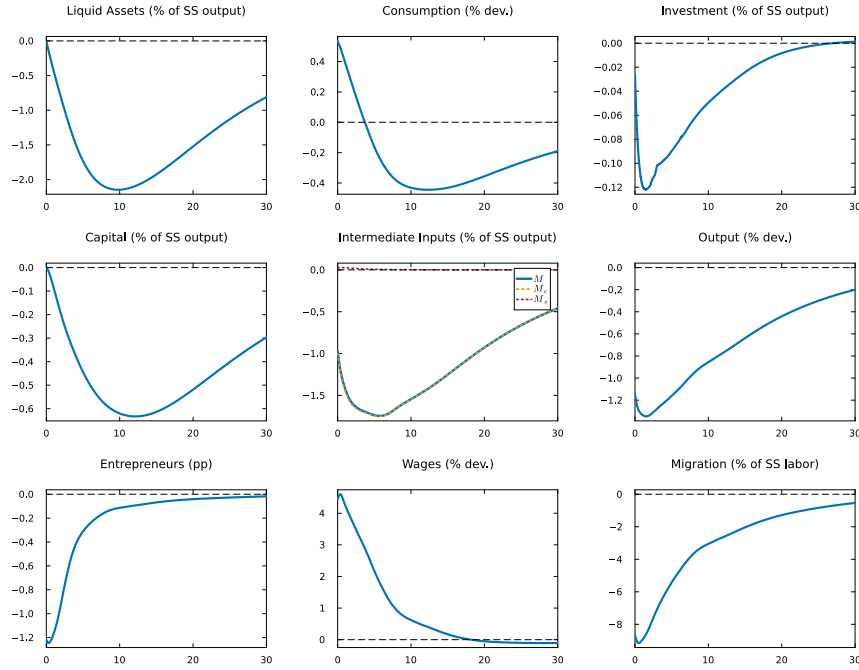


Figure 3: Aggregate Dynamics: Untreated region

Notes: The figure shows dynamic responses of aggregate variables in Region 2, following the cash transfer in region 1, under the partially integrated benchmark. Liquid wealth, investment, capital, and intermediate inputs are expressed as deviations from their steady-state values, normalized by steady-state output. Consumption, output, and wages are percent deviations from steady state. Entrepreneurship is reported in percentage points, and migration is measured as a percent of steady-state labor. The intermediate-input panel decomposes total intermediate spending into enterprise inputs, M_e , and subsistence agricultural inputs, M_s .

The wage increase has opposite effects on workers and entrepreneurs. For workers who remain in Region 2, higher wages raise labor income in the short run. But for current and potential entrepreneurs, the same wage increase makes business operation less attractive. It raises the cost of hired labor and increases the opportunity cost of entrepreneurship. As a result, entrepreneurship falls by about 1.2 percentage points. Enterprise input use

also declines sharply, while subsistence agricultural inputs remain essentially unchanged. Investment falls on impact, and the capital stock declines gradually as businesses scale down.

These forces generate a small but persistent contraction in Region 2 output. Consumption initially rises slightly, supported by higher wage income and by the decline in investment demand: as business opportunities become less attractive, households devote fewer resources to capital accumulation and temporarily divert some of those resources toward consumption. This gain is short-lived, however. As entrepreneurship, capital, enterprise inputs, and output fall, consumption turns negative relative to the no-transfer path. Liquid wealth also declines, reflecting the absence of a direct cash injection and the fall in business income. Over time, wages return toward steady state, migration flows unwind, and the production-side contraction gradually disappears.

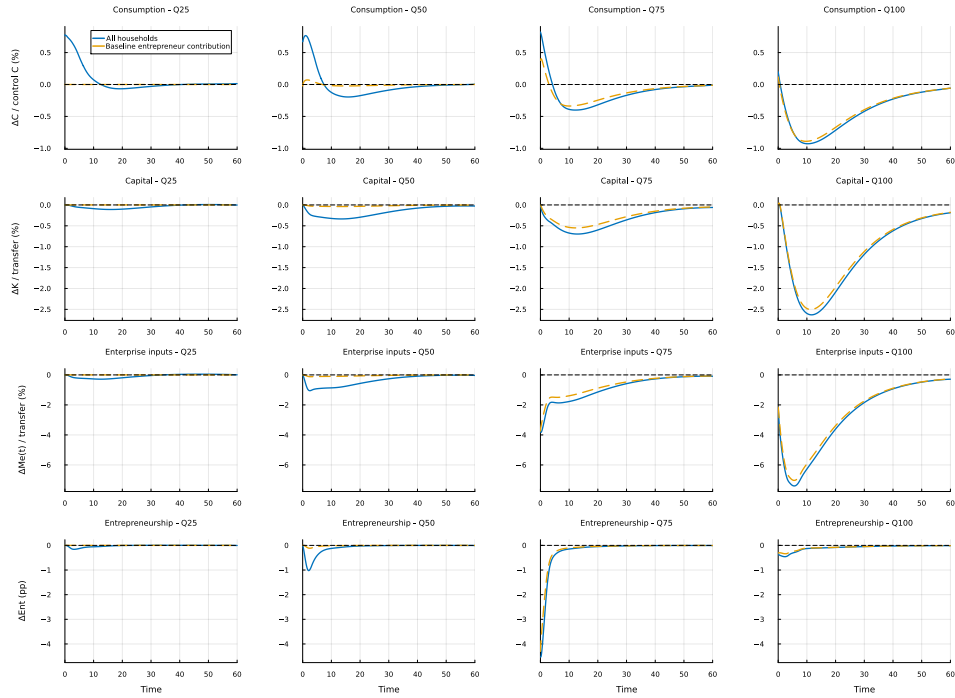


Figure 4: Untreated region Heterogeneous Responses

Notes: The graph shows responses by pre-transfer wealth quartile in Region 2. Blue lines show total quartile responses; yellow dashed lines show the contribution of baseline entrepreneurs. Each response is relative to the same group without the transfer. Region 2 does not receive the transfer, so responses reflect spillovers. Consumption is in percent; capital and enterprise inputs are percent of the transfer; entrepreneurship is in percentage points.

Figure 4 breaks the Region 2 spillover down by pre-transfer wealth quartile, revealing heterogeneity in who gains and who loses. The short-run consumption gains are concentrated in the lower and middle quartiles, consistent with these households benefiting from the temporary wage increase. In contrast, the top quartile experiences a sharper and more persistent consumption decline. This reflects its greater exposure to entrepreneurship and productive assets, whose returns fall as rising wages erode business profits.

The production response is concentrated higher in the wealth distribution. Capital and enterprise-input use fall only modestly in the bottom half of the distribution, but decline

substantially in the third and especially the top quartile. The dashed yellow lines show that much of this contraction comes from baseline entrepreneurs. Entrepreneurship falls most strongly in the third quartile, suggesting that some households near the business-operation margin exit or stop operating when wages rise. In the top quartile, the response is less about entry and exit and more about intensive-margin contraction: baseline entrepreneurs reduce inputs and capital as profits are compressed.

Taking Figures 3 and 4 together, the Region 2 results show a wage-positive but business-negative spillover. The treated-region boom draws labor away from the untreated region, temporarily raising wages for remaining workers but reducing profitability for entrepreneurs. Relative to Region 1, however, these effects are quantitatively small. The aggregate effects of the transfer are therefore dominated by the direct expansion in the treated region, while Region 2 absorbs a modest negative production spillover through the partially integrated labor market.

6. Welfare, Wage Effects, and Spatial Integration

In this section, we examine the welfare effects of the transfer and how the gains are distributed across households. We then study how the results change when prices (wages) are held fixed after the transfer. This exercise helps isolate the role of the general-equilibrium wage channel by removing the wage adjustments induced by the transfer. Finally, we consider alternative degrees of labor-market integration across regions and show how it shapes both aggregate responses and welfare.

We summarize welfare using a consumption-equivalent variation (CEV). Let $V_{ss}(s)$ denote the household value function in the pre-transfer stationary equilibrium, and let G_{ss} denote the associated stationary distribution over states s . Let G_{Tr} denote the distribution immediately after the transfer, that is, after liquid wealth has been shifted up by the transfer but before the subsequent transition has unfolded. Under scenario $i \in \{\text{Fixed Prices, Segmented, Benchmark, Full}\}$, let $V_0^i(s)$ denote the value at date 0 for a household that starts from post-transfer state s and then faces the entire future transition implied by scenario i . The aggregate consumption-equivalent welfare gain is

$$\omega_i = \left(\frac{\int V_0^i(s) dG_{Tr}(s)}{\int V_{ss}(s) dG_{ss}(s)} \right)^{\frac{1}{1-\sigma}} - 1.$$

This measure is the uniform proportional increase in consumption, relative to the stationary benchmark, that makes households indifferent between remaining in the pre-transfer stationary equilibrium and receiving the transfer under scenario i . In the fixed-price case, $V_0^i(\cdot) = V_{ss}(\cdot)$ because households face the stationary price environment after the transfer. Welfare therefore differs from the benchmark only because the transfer shifts the distribution from G_{ss} to G_{Tr} . In the remaining scenarios, V_0^i also incorporates the endogenous price path generated by the transfer.

To study heterogeneity, we compute the same object within baseline groups g , such as non-entrepreneurs versus entrepreneurs or baseline wealth quartiles. Let G_{ss}^g denote the stationary measure restricted to households in group g , and let G_{Tr}^g denote the post-transfer

measure. We define

$$\omega_i(g) = \left(\frac{\int V_0^i(s) dG_{Tr}^g(s)}{\int V_{ss}(s) dG_{ss}^g(s)} \right)^{\frac{1}{1-\sigma}} - 1.$$

This is a group-level ex-ante CEV. It therefore measures the uniform proportional change in baseline consumption that would make households in group g , before resolving their exact state within the group, indifferent to receiving the transfer under scenario i .

Table 6: Welfare Gains: Treated Region (%)

	Fixed Prices	Integration		
		Segmented	Benchmark	Full
Aggregate	10.03	10.54	10.37	10.21
<i>Baseline occupation</i>				
Entrepreneurs	10.32	9.15	9.45	9.70
Non-Entrepreneurs	9.90	11.20	10.80	10.45
<i>Baseline wealth quartile</i>				
Bottom 25%	9.85	11.51	11.01	10.57
25–50%	10.85	11.99	11.63	11.31
50–75%	11.62	11.79	11.71	11.63
Top 25%	7.37	6.15	6.48	6.74

6.1. Partial vs. General Equilibrium

Figure 5 compares the benchmark response in Region 1 to a partial-equilibrium counterfactual in which local wages are held fixed at their pre-transfer levels. The consumption and liquid-wealth responses are similar on impact, but the paths diverge over time. Under fixed prices, liquid wealth and consumption remain higher for longer because the transfer generates a stronger production response and therefore higher subsequent income. The main differences are on the production side. With wages fixed, entrepreneurship rises by more, capital accumulates more strongly and persistently, and enterprise input use remains elevated for much longer. Output also rises substantially more than in the benchmark equilibrium.

This comparison shows how general-equilibrium wage adjustment attenuates the productive response to the transfer. In the fixed-price counterfactual, firms can expand without bidding up labor costs, and households can shift into entrepreneurship without facing a higher wage outside option. In the benchmark equilibrium, by contrast, the transfer raises labor demand, pushing local wages up. Higher wages benefit workers, but they also compress entrepreneurial profits and make wage work more attractive for marginal households. This feedback reduces entry, slows capital accumulation, and dampens the rise in output. The fixed-price exercise therefore overstates the medium-run production effects of the transfer by shutting down the wage response that would normally ration local labor demand.

Table 6 shows that these quantity differences do not translate into large differences in aggregate welfare. Aggregate welfare gains are 10.03 percent under fixed prices and 10.37 percent in the benchmark. The reason is that wage adjustment changes the incidence of the gains

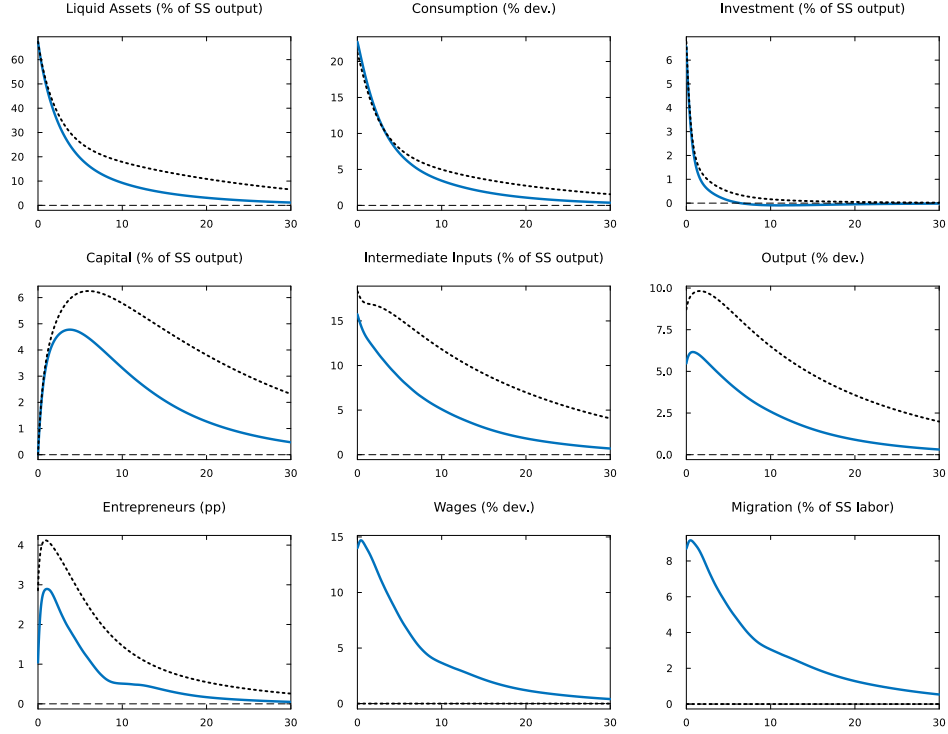


Figure 5: Aggregate Dynamics: Fixed Wage versus Benchmark

Notes: The figure shows the dynamic responses of aggregate variables in Region 1, the cash-transfer region, under the partially integrated benchmark economy and the fixed-wage economy. The blue line corresponds to the benchmark, while the black dotted line corresponds to the fixed-wage case. Liquid wealth, investment, capital, and intermediate inputs are expressed as deviations from their steady-state values, normalized by steady-state output. Consumption, output, and wages are reported as percent deviations from steady state. Entrepreneurship is reported in percentage points, and migration is measured as a percent of steady-state labor.

more than the aggregate value of the transfer. Fixed prices favor entrepreneurs, who benefit from stronger business expansion and unchanged labor costs: baseline entrepreneurs gain 10.32 percent under fixed prices, compared with 9.45 percent in the benchmark. The benchmark equilibrium instead shifts gains toward non-entrepreneurs and lower-wealth households through higher wages. Non-entrepreneurs gain 10.80 percent in the benchmark compared with 9.90 percent under fixed prices, and the bottom quartile gains 11.01 percent rather than 9.85 percent. Thus, the wage response dampens the production side of the local boom but redistributes welfare toward households that rely more on labor income.

6.2. Alternative Spatial Spillovers

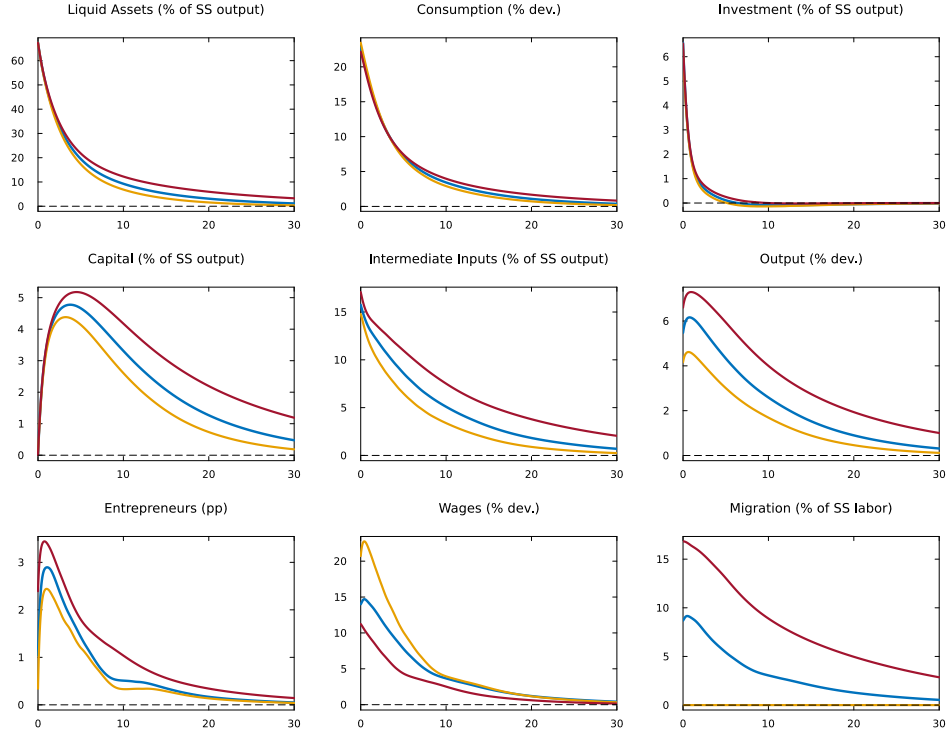


Figure 6: Aggregate Dynamics with Partial, Full, and Zero Labor-Market Integration

Notes: The figure compares the dynamic responses of the main aggregate variables in Region 1 under three labor-market integration regimes. The solid blue line corresponds to partial integration, the red line to full integration, and the yellow line to complete segmentation. Liquid wealth, investment, capital, and intermediate inputs are expressed as deviations from their steady-state values, normalized by steady-state output. Entrepreneurship is reported in percentage points, and migration is measured as a percent of steady-state labor.

Figure 6 compares the Region 1 response under three labor-market regimes: zero integration, the benchmark with partial integration, and full integration. The initial liquid-wealth and consumption responses are similar across regimes because households receive the same transfer. Over time, however, the paths diverge. As labor-market integration increases, liquid wealth and consumption remain higher for longer, reflecting a stronger production response. The main differences are on the production side: full integration generates the largest and most persistent increases in entrepreneurship, capital, enterprise-input use, and

output, while segmentation generates the weakest production expansion. The benchmark lies between these two cases.

The mechanism is the elasticity of local labor supply. Under segmentation, the transfer-induced increase in labor demand must be absorbed entirely within Region 1. Since workers cannot flow in from Region 2, wages rise sharply. This benefits local workers, but it also raises firms' labor costs and increases the wage outside option for households near the entrepreneurship margin. As a result, entry, input demand, and capital accumulation are muted. Under full integration, by contrast, labor can move more freely into Region 1. Migration absorbs more of the labor-demand shock, wage pressure is smaller, and firms expand at lower cost. Greater integration therefore shifts the response away from wages and toward quantities: more entrepreneurship, more capital, more intermediate-input use, and higher output.

Table 6 shows that these large differences in quantities do not translate into large differences in aggregate welfare for Region 1 as a whole. Welfare is highest under segmentation, at 10.54 percent, followed by the benchmark at 10.37 percent and full integration at 10.21 percent. The incidence, however, changes systematically. Segmentation generates the largest wage increase and therefore benefits non-entrepreneurs and lower-wealth households the most. Full integration dampens the wage response and allows stronger business expansion, shifting gains toward entrepreneurs and wealthier households. Thus, greater labor-market integration amplifies the production response to the transfer, but weakens the redistributive wage channel that benefits poorer and more labor-dependent households.

7. Domestically Financed Intervention

Our analysis so far treats the transfer as an exogenous, donor-funded inflow with no domestic cost. Increasingly, however, cash transfers are a domestic policy instrument in low-income countries, and a program operated at scale must be paid for out of the economy's own resources. We therefore ask how this transfer's effects change when the economy bears its cost. Specifically, we consider financing a one-time transfer with a permanent uniform lump-sum tax. We then examine how the benefits change relative to the no-financing case.

The experiment we consider in this section is the following. We abstract from the two-region structure and study a one-time transfer to the whole economy. Each household receives a liquid transfer Tr at $t = 0$. The government finances the transfer by borrowing at a constant real interest rate r_g . We set $r_g = 2\%$ in the baseline, which can be interpreted as a low-cost public borrowing benchmark, such as concessional or external financing. Note that this rate differs from the return on the household liquid asset, which is assumed to be lower reflecting domestic intermediation costs.

After the initial transfer, the government levies a permanent lump-sum tax that services the debt and keeps the debt stock constant. With population mass normalized to one, government debt per household is Tr , so the required per-household tax is

$$\tau_l = r_g Tr$$

For $t > 0$, the household budget constraint is therefore modified by subtracting the constant flow tax τ_l .

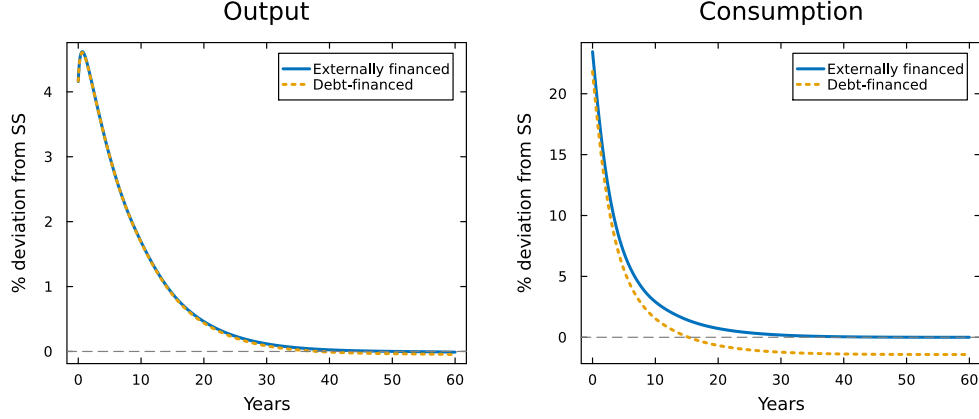


Figure 7: Output and Consumption: Transition Dynamics

Notes: The figure shows the transition dynamics of output and consumption under the externally financed and debt-financed transfers. Both variables are expressed as percent deviations from their initial stationary-equilibrium values in the baseline economy.

The main results for this experiment are reported in Figure 7. Relative to the externally financed benchmark, debt-service financing has almost no effect on the production side. This is visible in the output panel, where the externally financed and debt-financed paths are nearly indistinguishable. The same pattern holds for other production side variables, including intermediate input use and capital accumulation, which are not shown. The financing burden instead shows up primarily in consumption. In the long run, consumption is about 1.4% lower under debt financing, approximately matching the size of the permanent debt-service tax as a share of steady-state consumption.

To understand how financing affects welfare, Table 7 reports consumption-equivalent welfare gains under external and internal financing. The first column is, by construction, identical to the completely segmented labor-market case reported in Table 6. Relative to this externally financed benchmark, aggregate welfare falls by 1.65 percentage points under internal financing. The decline is broadly similar across wealth quartiles: the per-period tax required to service the debt is small, and for most households its welfare cost does not overturn the value of the upfront transfer. Welfare therefore remains positive throughout, and every quartile prefers the internally financed transfer to no transfer at all, including the top quartile, which gains least from the transfer's added liquidity. Only a small fraction at the very top lose from the internally financed transfer. These households are closest to the Ricardian benchmark: because they are far from liquidity constraints, the transfer provides little value, while the capitalized burden of the permanent tax outweighs the upfront payment.

7.1. Bang for Buck

In this subsection, we turn to a different welfare measure and discuss how the type of financing erodes or enhances the impact of cash transfers. We first define wealth-equivalent variation, then introduce a "bang-for-buck" statistic — the wealth-equivalent variation multiplier, to capture how a transfer is magnified or dampened across environments.

Table 7: Welfare Gains under External and Internal Financing

	Externally Financed	Internally Financed
Aggregate	10.54	8.89
<i>Baseline wealth quartile</i>		
Bottom 25%	11.51	9.68
25–50%	11.99	10.19
50–75%	11.79	10.10
Top 25%	6.15	4.91
<i>Negative CEV</i>		
% of households	1.52	2.21

Wealth-equivalent variation. We define this measure following [Conesa et al. \(2018\)](#). Let V^0 be the household value function in the baseline economy and V^p the value function under policy environment p . For a household in baseline state $s = (b, k, z, x)$, the *wealth-equivalent variation* $WEV_p(s)$ is the increment to baseline liquid wealth that leaves the household indifferent between remaining in the baseline economy and receiving the transfer Tr under environment p :

$$V^0(b + WEV_p(s), k, z, x) = V^p(b + Tr, k, z, x), \quad b + WEV_p(s) \geq 0, \quad (25)$$

where Tr is the cash transfer. $WEV_p(s) = Tr$ when environment p coincides with the baseline (i.e., no price effects), $WEV_p(s) > Tr$ when the general-equilibrium effects during transition leave the household better off than the transfer alone, and $WEV_p(s) < Tr$ when they leave it worse off. We summarize welfare by the population average

$$\overline{WEV}_p = \frac{\int WEV_p(s) dG^0(s)}{\int dG^0(s)}, \quad (26)$$

the mean wealth-equivalent variation across the baseline distribution G^0 .

We measure the benefit the economy receives relative to the cost of the transfer as $\overline{WEV}_p/Tr$. By definition, this ratio equals 1 when there are no price changes: a \$1 transfer makes the household exactly \$1 better off. Once price and tax responses enter, the ratio departs from 1. The higher it is, the larger the return over and above the private value of Tr . We summarize this economy-wide value of the policy with the *WEV multiplier*,

$$\mathcal{M}_p^{WEV} \equiv \frac{\overline{WEV}_p}{Tr}. \quad (27)$$

Table 8 reports $\mathcal{M}_p^{WEV}$ for the three cases. The partial-equilibrium case, as explained above, equals 1. The second case is a donor-funded transfer, which enters at no domestic cost and isolates the general-equilibrium amplification; here the multiplier rises by 2%, driven by wage increases that benefit the majority of households. The third case is a domestically

Table 8: **Wealth-Equivalent Variation Multiplier**

Case	$\mathcal{M}^{WEV}$	Mechanism captured
Partial-equilibrium cash transfer	1	Direct value of liquid transfer
Externally financed transfer	1.02	Direct value + GE effects
Debt-financed transfer	0.83	Direct value + GE effects – debt-service tax burden

financed transfer funded by a lump-sum tax. The multiplier falls to about 0.83, so domestic financing absorbs roughly 17% of the transfer’s wealth-equivalent value.

We interpret these exercises as simple accounting benchmarks that help quantify the role of financing, rather than as a complete treatment of optimal fiscal design. The donor-funded case is a natural upper benchmark in the sense that the domestic economy does not bear the fiscal cost of the transfer. The lump-sum-tax case anchors the lower end of the achievable range: because the tax burden is spread uniformly across households, including those with high marginal utility of liquid wealth, it erodes more welfare than a financing scheme targeted toward households with lower marginal utility. A more progressive or better-targeted domestic financing scheme would therefore likely lie between these two cases.

8. Conclusion

What are the macroeconomic and distributional effects of large cash transfers when implemented at scale? We provide the answer to this question from the perspective of a two-region, small open economy heterogeneous-agent model with occupational choice, financial and real frictions, and labor mobility across regions. Transfers are expected to increase consumption, productive expenditures in capital, and, more significantly, intermediate inputs. Consumption increases sharply and then gradually decreases, while productive capacity increases with a lag and remains elevated for longer due to irreversibility. Importantly, the effect will be substantially heterogeneous within and across regions: lower-wealth households exhibit the largest consumption response, while the richer entrepreneurs account for much of the expansion in capital and intermediate input use. Workers in non-treated regions benefit, but entrepreneurs there are negatively affected. The effects are predicted to gradually fade out as the economy converges back to the initial equilibrium.

Our model incorporates many prominent features of small open rural economies. Nevertheless, for computational tractability, we abstract from several potentially important margins that provide promising avenues for future research.

The entrepreneurship in our model are intended to represent farms producing cash crops, small manufacturing establishments, and retail businesses. An important distinction that we abstract from is that some of these activities produce tradable goods, whereas others produce primarily non-tradable goods and services. In a model with both tradable and non-tradable sectors, transfers would likely generate a temporary appreciation of the local real exchange rate through an increase in the relative price of non-tradables. This price pressure would be partially offset by the expansion in productive capacity resulting from higher capital accumulation and the increased use of intermediate inputs. Nevertheless, the

net effect is likely to be positive, since in our quantitative exercises, consumption increases by about 20 percent while output rises by only 6 percent.

We model consumption as non-durable, thereby abstracting from an important mechanism through which households may spread the benefits of transfers over time: the accumulation of durable goods and housing investment. This would be a particularly important margin in an environment with low returns to saving. To the extent that housing investment relies on local labor inputs, incorporating this margin would likely amplify the equilibrium hike in wages.

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